

THE POTENTIAL DISTRIBUTION AND MORPHOLOGICAL DATA OF *Podarcis siculus* (RAFINESQUE-SCHMALSTZ, 1810) WITH NEW LOCALITY RECORDS FROM TURKEY (SQUAMATA: SAURIA: LACERTIDAE)

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Submitted March 23, 2017

In the present study, we contributed to knowledge of distribution of the Italian wall lizard, *P. siculus* in Turkey. Besides, we gave some morphological data about species from Akçakoca population. The morphological data of the specimens from Akçakoca were generally found to be similar with the known literature. Only some morphological characteristics were observed to be different from the features given in the literature. The mean of supraciliar granules (SCG) was found lower and the mean of median gularia (MG) was found to be higher than the mean values given in the literature. In addition, we found high percentage of five supralabial plates in anterior of oculars. The distribution of the species was found to be restricted between northeastern Marmara and western Black Sea regions, according to our distributional model. On the other hand, the distributional data of *P. siculus* in the present study exhibited that Çanakkale and Samsun have not suitable habitats for the species. High AUC and lower standard deviation values (0.986 ± 0.006) showed that the potential distribution area of the species was congruent with its habitat preferences inhabited localities.

Keywords: habitat preference; morphology; Italian wall lizard; distributional data; maximum entropy.

INTRODUCTION

Podarcis (Wagler, 1830) genus is a monophyletic lizard group with full recognized twenty four species [*P. bocagei* (Seoane, 1884); *P. carbonelli* (Perez-Mellado, 1981); *P. cretensis* (Wettstein, 1952); *P. erhardii* Bedriaga, 1882); *P. filfolensis* Bedriaga, 1876; *P. gaigeae* (Werner, 1930); *P. guadarramae* (Bosca, 1916); *P. hispanicus* (Steindachner, 1870); *P. ionicus* (Lehrs, 1902); *P. leventidis* Lymberakis et al., 2008; *P. lilfordi* (Günther, 1874); *P. liolepis* (Boulenger, 1905); *P. melisellensis* (Braun, 1877); *P. milensis* (Bedriaga, 1882); *P. muralis* (Laurenti, 1768); *P. peloponnesiacus* (Bibron et Bory, 1833); *P. pityusensis* (Bosca, 1883); *P. raffonei* (Mertens, 1952); *P. siculus* (Rafinesque-Schmalstz, 1810); *P. tauricus* (Pallas, 1814); *P. tilingueta* (Gmelin, 1789); *P. vaucheri* (Boulenger, 1905); *P. virens* Geniez et al., 2014; and *P. waglerianus* (Gistel, 1868)] in the Lacertidae family (Arnold, 1973; Oliverio et al., 2000; Carranza et al., 2004; Harris, 1999; Harris et al., 2005; Sindaco and Jeremcenko, 2008; Sindaco et al., 2013; Psonis et al.,

2016). The genus has a great distribution area in the western Palearctic region (eastern Asia, southern Europe and northern Africa) and it is known as a dominant group in the Mediterranean region (Arnold, 1973; Harris and Arnold, 1999; Poulakakis et al., 2003; Ananjeva et al., 2006; Arnold et al., 2007; Kaliontzopoulou et al., 2008; Sindaco and Jeremcenko, 2008). Only three *Podarcis* species (*P. muralis*, *P. siculus*, and *P. tauricus*) of the genus are distributed in Turkey (Başoğlu and Baran, 1977; Baran and Atatür, 1998; Sindaco et al., 2000; Baran et al., 2013) while all species of the genus have almost occurred in the Continental Europe (<http://www.lacerta.de>; <http://www.reptile-database.org>).

The Italian wall lizard, *P. siculus* is a polytypic species distributed mainly in Italian Peninsula (the islands of Sicilia and Sardinia, and many islands in the Tyrrhenian Sea) and other locations, including the Adriatic coasts from Slovenia to Montenegro, edge of southern Switzerland, Corsica in southeastern France, Iberian Peninsula, Albania, Montenegro, northern and northwestern Turkey, Libya, Tunisia, and United States in the world (Behler and King, 1979; Conant and Collins, 1991; Gasc et al., 1997; Podnar et al., 2005; Mollov, 2009; Rivera et al., 2011; Jablonski and Stloukal, 2012; Ilgaz et al., 2013;

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Kolbe et al., 2013; Silva-Rocha et al., 2014; Sillero et al., 2014; Tok et al., 2015; Mizsei et al., 2016). Although the species has 52 morphological and biogeographical subspecies (Henle and Klaver, 1986), more than twenty subspecies have been currently accepted phylogenetically based on mitochondrial DNA sequences (Podnar et al., 2005).

The first Turkish record of Italian wall lizard was given from Anatolian part of İstanbul province and the specimens of species were morphologically described as the representatives of *P. s. hieroglyphicus* by Berhold (1942). Later, many records were given from İstanbul and islands of Marmara (Bird, 1936; Bodenheimer, 1944; Mertens, 1960; Clark and Clark, 1973; Baçoğlu and Baran, 1977; Franzen, 1990; Çevik, 1999; Jablonski and Stloukal, 2012), Bursa (Uğurtaş et al., 2000; Mollov, 2009; Arslan et al., 2013), Çanakkale (Hür et al., 2008; Tok and Çiçek, 2014; Tok et al., 2015), Zonguldak (İlgaz et al., 2013) and Samsun (Tok et al., 2015). According to current literature, *P. s. siculus* is only found in Turkey (<http://www.lacerta.de>; Silva-Rocha et al. 2014).

Podarcis siculus is a medium-sized lizard among the Lacertidae family. The habitat of species occurs in the walls of buildings, gardens and cemeteries within cities. Moreover, the species prefers to live rocky and stony areas in the coastal regions. The woody areas in the city parks are other preferred habitats of the species (Baran and Atatür, 1998; Çevik, 1999; Baran et al., 2013). The species is distributed from the sea level to about 1800 m elevations, including suitable habitats of the populations (Baran et al., 2013).

Many different species distribution modeling methods are employed to determine potential suitable habitats of a known species (Guisan and Zimmermann, 2000; Guisan and Thuiller, 2005; Elith et al., 2006; Guisan et al., 2007; Wisz et al., 2008; Kumar and Stohlgren, 2009). Ecological Niche Modeling (ENM) is an important tool widely used to predict the potential geographic ranges of a species in recent years by many researchers (Varela et al., 2014; Wigra and Majtyca, 2015; Gül et al., 2015, 2016; Kurnaz et al., 2016; Sanchooli, 2016; Yousefkhani et al., 2016a). Learning where animals live in a region and predicting which climatic factor affects their habitat preferences are important for conservation and ecology of them (Kumar and Stohlgren, 2009; Gül, 2013; Kurnaz et al., 2016). Based on the maximum-entropy (MaxEnt ver. 3.3.3k jar Software; Phillips et al., 2006) approach for species habitat modeling, this software program receives as input a string of environmental (such as elevation, precipitation, etc.) variables as predictors, as well as a string of georeferenced locations, and produces a model of the range of a species (Sanchooli, 2016). This approach was used in many studies for amphibian and

reptilian species from Turkey (Fattahi et al., 2014; Gül, 2015; Gül et al., 2015, 2016; Afsar et al., 2016; Kurnaz et al., 2016; Tok et al., 2016; Yousefkhani et al., 2016a, 2016b).

In the present study, we performed to determine the potential geographic distribution and climatic preferences of the species, *P. siculus* using more new locality records in Turkey. Also, we presented some morphological data in a population of the species from Düzce (vicinity of Akçakoca) province which is a new locality for the distribution of the species in Turkey. Moreover, we filled some blank locations in the distribution area for the species in Turkey. Moreover, we present that climatic conditions in the potential distribution area of the species are congruent with its habitat preferences inhabited localities.

MATERIAL AND METHODS

Geographical occurrence data of species. During a three-year (2013 – 2016) field survey for the amphibian and reptilian species in Turkey, a total of 33 locality records for *P. siculus* was collected from eight different provinces (Samsun, Zonguldak, Düzce, Sakarya, Kocaeli, İstanbul, Bursa, and Çanakkale) in the distribution area of the species. In addition, all published literature on the species in Turkey were investigated and 15 occurrence geographical data were accumulated (Bird, 1936; Bodenheimer, 1944; Baçoğlu and Baran, 1977; Çevik, 1999; Uğurtaş et al., 2000; Hür et al., 2008; Mollov, 2009; Jablonski and Stloukal, 2012; Arslan et al., 2013; İlgaz et al., 2013; Tok and Çiçek, 2014; Tok et al., 2015). All locality records used in the present study are given in Table 1.

Morphological data. Five specimens (2 ♂♂, 3 ♀♀) were collected from a locality; vicinity of Akçakoca in Düzce province (on 21 July 2016, GPS data, 41°05.654' N 31°11.592' E, 4 m a.s.l.). All specimens were anesthetized with ether, fixed with a 10% formaldehyde injection, and deposited in 70% ethanol. They were deposited in the Zoology Lab. (collection number: KZL-238 for Akçakoca) of the Department of Biology at the Faculty of Science, Karadeniz Technical University.

Mensural, meristic and qualitative data were recorded following the systems of Çevik (1999), İlgaz et al. (2013), and Tok et al. (2015). All pholidolial characters were examined under a stereo microscope and all specimens' morphometric features were measured using a digital caliper. All measured data were compared with the studies of Çevik (1999), İlgaz et al. (2013), and Tok et al. (2015). The following pholidolial characteristics were evaluated: supraciliar granules (right-left, SCGa-SCGb),

TABLE 1. The Geopoint Data of *Podarcis siculus* in Turkey

No.	Latitude	Longitude	Location	Reference
1	40.967491	29.146893	İstanbul	Bodenheimer, 1944
2	41.145593	29.178306	Beykoz/İstanbul	Bird, 1936
3	41.018811	28.964452	Bosphorus/İstanbul	Bodenheimer, 1944
4	40.189232	29.061353	Bursa	Uğurtaş and Yıldırımhan, 2000
5	40.469180	29.669197	İznik/Bursa	Uğurtaş and Yıldırımhan, 2000
6	39.857782	27.252364	Hamdibey/Çanakkale	Hür et al., 2008
7	39.805281	26.930016	Çanakkale	Tok and Çiçek, 2014
8	39.807066	27.043324	Çanakkale	Tok and Çiçek, 2014
9	39.814654	27.112862	Çanakkale	Tok and Çiçek, 2014
10	41.332474	36.286102	Atakum/Samsun	Tok et al., 2015
11	40.408188	26.663484	Gelibolu/Çanakkale	Tok et al., 2015
12	40.364672	28.908183	Mudanya/Bursa	Mollov, 2009
13	40.590348	27.602614	Marmara Islands	Bodenheimer, 1944
14	41.562205	32.022630	Filyos/Zonguldak	Ilgaz et al., 2013
15	40.977406	29.043217	İstanbul	Jablonski and Stloukal, 2012
16	40.788783	29.382767	Darıca/Kocaeli	This study
17	40.738450	30.297350	Serdivan/Sakarya	This study
18	40.713700	30.354083	Arifiye/Sakarya	This study
19	40.703750	30.139117	Maşukiye/Kocaeli	This study
20	40.697367	30.165917	Sapanca/Sakarya	This study
21	40.713017	30.334017	Arifiye/Sakarya	This study
22	40.702567	30.144150	Maşukiye/Kocaeli	This study
23	41.270300	31.449567	Ereğli/Zonguldak	This study
24	41.227450	31.960783	Devrek/Zonguldak	This study
25	41.559168	32.029819	Filyos/Zonguldak	This study
26	41.418817	32.092150	Çaycuma/Zonguldak	This study
27	41.327884	36.279319	Atakum/Samsun	This study
28	40.416186	26.677902	Gelibolu/Çanakkale	This study
29	41.430200	32.088050	Çaycuma/Zonguldak	This study
30	41.481333	31.836967	Kilimli/Zonguldak	This study
31	41.570267	32.037267	Filyos/Zonguldak	This study
32	41.568200	32.041517	Filyos/Zonguldak	This study
33	41.581175	32.081397	Filyos/Zonguldak	This study
34	41.553200	32.060033	Filyos/Zonguldak	This study
35	41.095900	31.193200	Akçakoca/Düzce	This study
36	41.261117	31.440883	Ereğli/Zonguldak	This study
37	41.261733	31.480467	Ereğli/Zonguldak	This study
38	41.548900	32.077200	Filyos/Zonguldak	This study
39	41.024062	28.901754	Güngören/İstanbul	This study
40	41.014613	29.059118	Üsküdar/İstanbul	This study
41	40.883121	29.361183	Tuzla/İstanbul	This study
42	41.033931	28.977754	Beyoğlu/İstanbul	This study
43	40.987198	29.238861	Samandıra/İstanbul	This study
44	40.203879	29.066435	Bursa	This study
45	40.784554	29.408149	Gebze/Kocaeli	This study
46	40.822835	29.432649	Gebze/Kocaeli	This study
47	40.493853	30.281678	Sakarya	This study
48	40.818874	30.414328	Adapazarı/Sakarya	This study



Fig. 1. The habitat of *Podarcis siculus* in Akçakoca.

loreal plates back of postnasal plates and front of preocular plates (right-left, LOa-LOb), supraciliar plates (right-left, SCPa-SCPb), supralabial plates (right-left, SRLa-SRLb, number of labials anterior to center of eye), sublabial plates (right-left, SLPa-SLPb), inframaxillar plates (right-left, IMa-IMb), transversal series of gular scales between inframaxillar symphysis and collar (MG), collar (C), supratemporals (right-left, STa-STb), ventral plates (transversal and longitudinal, TVP and LVP), femoral pores (right-left, FPa-FPb), subdigital lamellae in the 4th toe (right-left, SDLa-SDLb), transversal series of dorsal scales at the midbody (DS), and number of preanal scales surrounding anals (PA1) and all plates surrounding anals (PA2).

The morphometric measurements in this study following: snout-vent length (SVL), tip of snout to anal cleft; tail length (TL), anal cleft to tip of tail; pileus width (PW), at widest point between parietal plates; pileus length (PL), tip of snout to posterior margins of parietals; head width (HW), at widest point of head; head length (HL), tip of snout to posterior margin of ear opening; total body length (TBL), tip of snout to tip of tail.

The habitat of the specimens from Akçakoca consists of rocky areas near the Düzce-Zonguldak highway (Fig. 1). *P. siculus* and *Lacerta viridis* (Laurenti, 1768) live in sympatry at the locality. The specimens were found during a day excursion between 12:30 and 14:30. The temperature was about 27 – 28°C.

Material. KZL-238/2016, 2 ♂♂, 3 ♀♀, 21.07.2016, Akçakoca, Düzce leg. U. Bülbül, M. Kurnaz, and A. İ. Eroğlu

Species distribution modeling. 19 bioclimatic variables were downloaded from Global Climate Data to construct species distribution modeling (Heikkinen et al.,

2006, available at www.worldclim.org). These data were generated from global esri grids in the highest resolution [30'' (~1 km)] for current conditions (~1950 – 2000). Each bioclimatic variable was limited the land border of Turkey using by Arc Toolbox (extract by mask) in ArcGIS ver. 10.3 software. In order to eliminate the negative effect may be resulted from other environmental variables, Pearson correlation coefficient ($0.75 < r < -0.75$) was performed using ENMTools 1.4 (Warren et al., 2010) for all variables, and highly correlated variables were left out of analyze for distribution modeling of *P. siculus*. After the correlation analyze, we chose six bioclimatic variables; Temperature seasonality (standard deviation $\times$ 100) (Bio 4), Mean temperature of wettest quarter (Bio 8), Mean temperature of coldest quarter (Bio 11), Precipitation of wettest month (Bio 13), Precipitation of driest month (Bio 14) and Precipitation seasonality (standard deviation of monthly precipitation) (Bio 15).

Maxent 3.3.3k (Philips et al., 2006) software was used to perform species distribution modeling. To develop the model, 48 occurrence data based on literature and new locality records were used. 25% of the occurrence data were set aside as test points, and 10.000 background points were used to determine the distribution. Additionally, the regularization multiplier = 0.5, maximum iterations = 500, and convergence threshold = 10^{-5} were chosen in Maxent. In order to test the variable importance, the jackknife test of variable importance was chosen in Maxent, and the model was run as ten replicates. The result of the receiver operating characteristic (ROC) curve is important for model sensitization and the value of the area under the curve (AUC) closest to 1 indicated the best model performance. A value near 0.5 suggests that the result is not better than random (Raes and ter Steege, 2007; Gallien et al., 2012).

RESULTS

Morphological data

Meristic features. There was no contact between rostral and internasal plates in the 5 specimens. The interparietal plates were in contact with the occipital plate in all specimens. The postnasal plate was single on each side in all specimens. The postnasal plate was single on each side in all specimens. The masseteric plate was present on each side of the head in all specimens. The row of supraciliar granules was always complete. SCPa was 7 (60%) in 3 specimens and 6 (40%) in 2 specimens. SCPb was 7 (40%) in 2 specimens, 6 (40%) in 2 specimens, and 5 (20%) in 1 specimen. LOs were always 2 on both sides

of the head in all specimens. A large and clear tympanicum was present on both sides of the head in all specimens. Four supraocular plates were present on both sides of the head in all specimens. SRLa was 5 (60%) in 3 specimens while it was 4 (40%) in 2 specimens. SRLb was 5 (80%) in 4 specimens and 4 (20%) in 1 specimen. SLPa and SLPb were 8 (100%) in all specimens. STs were large and narrow; the first one was longest in all specimens. STa was 8 (20%) in 1 specimen, 7 (60%) in 3 specimens and 6 (20%) in 1 specimen. STb was 7 (60%) in 3 specimens, 6 (20%) in 1 specimen, and 5 (20%) in 1 specimen. IMa was 6 (40%) in 2 specimens and 5 (60%) in 3 specimens. IMb was 7 (20%) in 1 specimen, 6 (20%) in 1 specimen and 5 (60%) in 3 specimens. The first 3 of IMs were in contact in all specimens. Collars were always smooth edged in all specimens. Dorsal body scales were small and smooth. Subdigital lamellae in the fourth toe were smooth. Anal plate was single in all specimens. PA1 and PA2 were 2 (100%) and 6 (100%) in all specimens, respectively.

Metric measurements. The maximum TBL for female specimens was 183.68 mm while it was 255.75 mm for male specimens. The maximum SVLs for females and males were 67.13 mm and 78.51 mm, respectively. In all specimens, the means of PL, PW, HL, and HW were 17.29 mm (range: 15.37 – 20.18), 7.75 mm (range: 6.97 – 8.84), 18.55 mm (range: 16.01 – 20.20), and 10.66 mm (range: 8.42 – 12.66), respectively. Descriptive statistics of pholidolial characteristics and morphometric measurements of the specimens collected from Akçakoca, Düzce are shown in Table 2.



Fig. 2. The general view of male *Podarcis siculus* specimen from Akçakoca.

Color-pattern features. The top of the head was brownish in both male and female specimens. The lateral sides of the head, especially temporal and supraciliary bands were light brown in females and grayish in males. The color of the supralabial plates varied from creamy to light brown in both sexes. The dorsum coloration was generally yellowish and greenish in males and green-brownish in females. The lateral side of the body was brown with dense, rounded green spots in both sexes. The ventral region was light greenish as in the lower side of the head in both sexes. In some cases, the outer ventral plates were covered with rounded light greenish or bluish spots (Fig. 2).

Distributional data

We added 33 new geopoint records that were not given in the literature. Kocaeli, Sakarya, and Düzce are new provinces for distribution of *P. siculus* in Turkey. Apart from the known locality (Filyos) in the literature,

TABLE 2. Descriptive Statistics of Some Pholidolial Characteristics and Morphometric Measurements of *Podarcis siculus* Collected from Akçakoca, Düzce

Characters	N	Mean	Min	Max	SE	Characters	N	Mean	Min	Max	SE
SCGa	5	7.6	5.0	11.0	1.08	STb	5	6.4	5.0	7.0	0.4
SCGb	5	8.8	7.0	11.0	0.74	TVP	5	28.0	27.0	29.0	0.4
SCPa	5	6.6	6.0	7.0	0.25	LVP	5	6.0	6.0	6.0	—
SCPb	5	6.2	5.0	7.0	0.37	FPa	5	24.6	24.0	25.0	0.3
SRLa	5	4.6	4.0	5.0	0.25	FPb	5	24.4	24.0	25.0	0.3
SRLb	5	4.8	4.0	5.0	0.20	DS	5	71.0	67.0	74.0	1.2
SLPa	5	8.0	8.0	8.0	—	PA1	5	2.0	2.0	2.0	—
SLPb	5	8.0	8.0	8.0	—	PA2	5	6.0	6.0	6.0	—
SDLa	5	30.6	29.0	31.0	0.40	SVL	5	71.7	67.7	77.5	2.3
SDLb	5	31.8	31.0	33.0	0.37	TL	3	130.0	94.9	180.0	25.7
IMa	5	5.4	5.0	6.0	0.25	PW	5	7.8	7.0	8.8	0.4
IMb	5	5.6	5.0	7.0	0.40	PL	5	17.3	15.4	20.2	1.1
MG	5	31.4	30.0	34.0	0.68	HW	5	10.7	8.4	12.7	0.8
C	5	10.2	8.0	12.0	0.80	HL	5	18.6	16.0	22.2	1.4
STa	5	7.0	6.0	8.0	0.32	TBL	3	200.7	162.8	255.8	28.2

For abbreviations, see the text; SE, standard error.

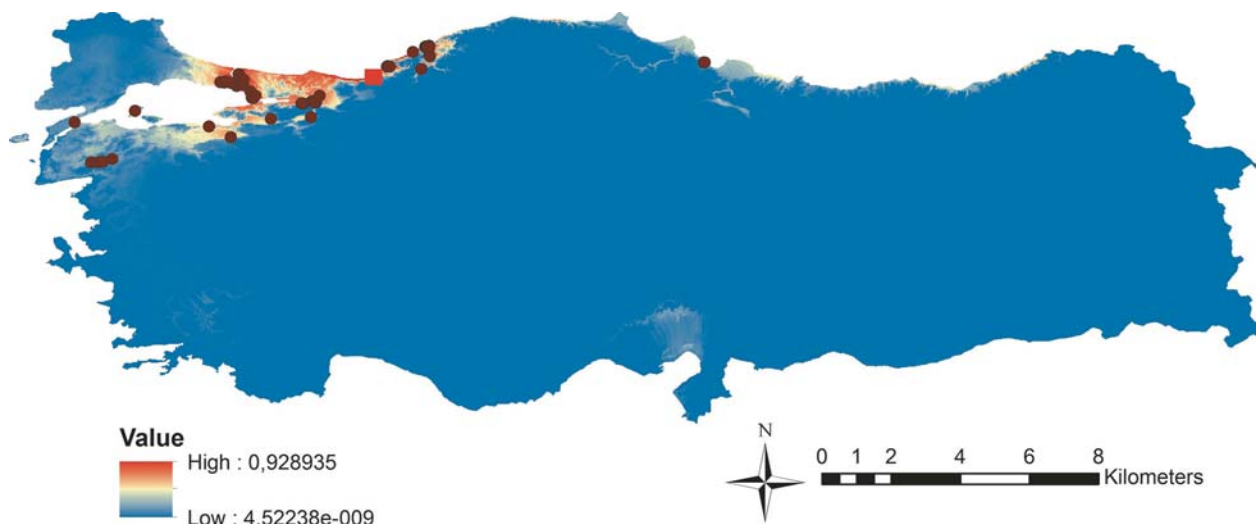


Fig. 3. The expected potential distribution of *Podarcis siculus* based on current climatic conditions.

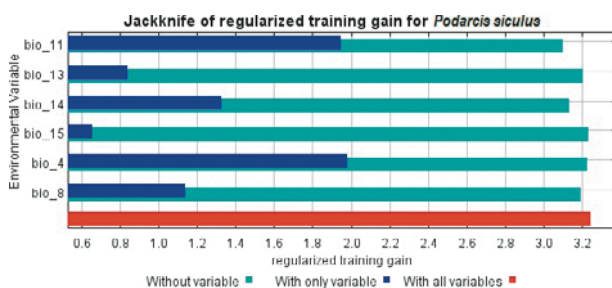


Fig. 4. The results of the jackknife test of variable importance used in the model for *Podarcis siculus*.

we presented 11 geopoint records for Zonguldak province (6 records from Çaycuma, 3 ones from Karadeniz Ereğli, 1 locality from Devrek and 1 record from Kilimli). According to our constructed model, the distribution of species was generally observed to be in the northeastern Marmara region and the coast of the western Black Sea region. In addition, our model shows that the species may be distributed in low level in the coast of middle and the eastern Black Sea region as well as Samsun province

which is known easternmost locality of *P. siculus* in Turkey. Moreover, the distribution model of the subspecies was well fit the distribution between İstanbul and Zonguldak provinces (red color), while it was not fully good fit the distribution between Zonguldak and Samsun provinces (grayish color) (Fig. 3). The environmental variables contributed to highest gain for distribution of *P. siculus* were Bio 4 (Temperature seasonality (standard deviation $\times$ 100), 41.1%), Bio 14 (Precipitation of driest month, 20.8%) and Bio 11 (Mean temperature of coldest quarter, 19.6%). Other variables showed a lower contribution than percentage 20 (Table 2). Furthermore, Bio 4 presented the most useful information when used by itself in the Maxent analyze. According to the jackknife of regularized training gain of the species, the performance of the model was the lowest without Bio 4 (Fig. 4). All bioclimatic variables used in the present study are given in Table 3 with the contribution values. The final constructed models resulted in good and high AUC with a low standard deviation (0.986 ± 0.006) for *P. siculus*. The high AUC result and lower standard deviation exhibit that potential distribution area of the species was congruent with its habitat preferences inhabited localities.

TABLE 3. The Contributions of the Climatic Conditions to Distribution Modeling of *Podarcis siculus*

Bioclimatic Variables	Percent Contribution, %
Bio 4	41.1
Bio 14	20.8
Bio 11	19.6
Bio 8	7.4
Bio 15	6.3
Bio 13	4.3

DISCUSSION

Morphological data. The morphological characteristics (pholidolial, color-pattern and morphometric) of the specimens in Akçakoca population were generally found to be similar to the specimens used in the studies in the literature (Bird, 1936; Çevik, 1999; Uğurtaş et al., 2000; Ilgaz et al., 2013; Tok et al., 2015). The studies re-

lated to Turkish populations of *P. siculus* were mainly based on morphological investigations (Berthold, 1842; Bird, 1936; Çevik, 1999; Uğurtaş et al., 2000; Ilgaz et al., 2013; Tok et al., 2015). When we compared our results from the Akçakoca population to the pholidolial records given in the literature related to the *P. siculus*; only several pholidolial characteristics were found to be different from the values given in the literature. The mean of supraciliar granules (SCG) was found lower and the mean of median gularia (MG) was found to be higher the mean values given in the literature. The comparison is given in Table 4. The supralabial plates (SRLs) in the anterior of the eyes are generally four in each side of the head except *Anatololacerta* group (species in this group have only five supralabial plates in the anterior of oculars). *P. siculus* has normally four supralabial plates according to morphological studies in the literature (Ilgaz et al., 2013; Tok et al., 2015). However, we found generally five supralabial plates on each side of the head for the specimens from Akçakoca population. Although our results were similar to the records given in the literature in terms of many morphological characteristics, the number of the specimens in our study was very low. More specimens should be investigated to evaluate the similarity of

the Akçakoca population to other Turkish populations of *P. siculus*.

Distributional data. *P. siculus* had a disconnected distribution for some localities in Turkey, according to all locality records (in the present study and literature). Consistent with this data, the distributional model in the present study showed that northeastern Marmara and western Black Sea region were highly suitable habitats for *P. siculus*. The species was also recorded in some localities from Samsun and Çanakkale provinces (Hür et al., 2008; Tok and Çiçek, 2014; Tok et al., 2015). However, the suitability of these localities in the distribution model was found lower in the present study.

According to the literature, *P. siculus* had very scattered populations in its distribution area except the Italian peninsula (Radovanović, 1956; Schneider, 1971; Gorman et al., 1975; Behler and King, 1979; Conant and Collins, 1991). This characteristic has presented that the species will be harmonized with different habitat structures and climatic variables. Consistent with this situation, ecological niche model results in the present study showed that different climatic variables (temperature, precipitation, drought, etc.) contribute to the distribution of the species in Turkey.

TABLE 4. Comparison of Some Pholidolial Characteristics and Morphometric Measurements of Our Specimens with Those Published

Reference		SCG	MG	C	DS	TVP			FP	SDL
						♀	♂	♀ + ♂		
Bird (1936)	<i>N</i>		6	6	6			6		6
	mean		28.33	10.67	69.67			27.83		27.67
	range		26 – 31	10 – 11	64 – 76			26 – 29		24 – 31
Çevik (1999)	<i>N</i>	42	41	42	43	14	18	42	42	42
	mean	9.2	29.4	11.1	71.8	28.8	26.0		24.1	32.6
	range	7 – 12	25 – 34	10 – 13	62 – 80	27 – 30	25 – 28		21 – 27	26 – 36
Uğurtaş et al. (2000)	<i>N</i>	6	6	6	6	2	4		6	6
	mean	10.2	28.0	11.8	71.5	28.5	26.5		24.7	31.2
	range	9 – 11	27 – 30	11 – 13	69 – 74	28 – 29	25 – 28		22 – 26	30 – 32
Ilgaz et al. (2013)	<i>N</i>	6	6	6	6	2	4	6	6	6
	mean	10.2	28.0	11.8	71.5	28.5	26.5	27.2	24.7	31.2
	range	9 – 11	27 – 30	11 – 13	69 – 71	28 – 29	25 – 28	27 – 29	22 – 26	30 – 32
Tok et al. (2015) Gelibolu	<i>N</i>	14	14	14	13	7	7	14	14	14
	mean	10.5	28.9	10.9	72.8	28 – 7	25.7	27.2	23.9	33.5
	range	8 – 13	26 – 31	10 – 12	69 – 77	27 – 30	25 – 27	25 – 30	22 – 25	30 – 37
Tok et al. (2015) Samsun	<i>N</i>	5	5	5	5			5	5	5
	mean	10.4	27.4	11.2	69.4			27.2	25.6	32.4
	range	8 – 12	26 – 30	10 – 12	66 – 72			26 – 29	24 – 27	32 – 34
This study	<i>N</i>	5	5	5	5			5	5	5
	mean	8.8	31.4	10.2	71.0			28.0	24.4	31.8
	range	7 – 11*	30 – 34	10 – 12	67 – 74			27 – 29	24 – 25*	31 – 33*

For abbreviations, see the text.

* The values are used for left side of the body.

Geographical distribution has an importance for conservation, ecology and evolution of the species (Kurnaz et al., 2016). It is an important tool to get an answer to some questions such as “How many clades occurs in the species and where are geographic limitations in the distribution of species?” Senczuk et al. (2017) interpreted at least two aspects of genetic complexity associated with different time-scale processes for *P. siculus*. The first was major geographic barriers to the elongated allopatric differentiation process during the Early Pleistocene and the second was an independent local differentiation in response to climatic fluctuation during the Early-Middle Pleistocene transition. Our climatic results have shown that allopatric distribution of *P. siculus* between Turkey and Europe earns different climatic preferences for the species. Mean diurnal range (Bio 2), temperature seasonality (Bio 4), minimum temperature of coldest month (Bio 6), temperature annual range (Bio 7), annual precipitation (Bio 12) and precipitation seasonality (Bio 15) contributed to European distribution according to study of Senczuk et al. (2017) while temperature seasonality (Bio 4), mean temperature of wettest quarter (Bio 8), mean temperature of coldest quarter (Bio 11), precipitation of wettest month (Bio 13), precipitation of driest month (Bio 14) and precipitation seasonality (standard deviation of monthly precipitation) (Bio 15) contributed to Turkish distribution in the present study. According to these data, it was seen that the distribution of the species depended on temperature and precipitation seasonality for both European and Turkish distributions. Other climatic variables exhibit geographical and habitat preference differentiations of the species between Turkish and European distributions.

CONCLUSION

The present study provides new locality records of *P. siculus* from Turkey and first comprehensive assessment on potential distribution of the species in Turkey using bioclimatic variables. According to the mean of supraciliar granules, the mean of median gularia and the supralabial plates (SRLs) in the anterior of the eyes, our studies population was different from other populations of *P. siculus* in Turkey. Phylogeny of *P. siculus* has not been investigated for Turkish populations. Based on their molecular data, Podnar et al. (2005) and Senczuk et al. (2017) stated that the Italian wall lizard was subdivided into two major clades correspond to formerly separated and differentiated population groups. Podnar et al. (2005) also explained that one of these groups could have entered the distribution areas of the other groups and hybridized with them. The allopatric geography between

Europe and Turkey constitutes major questions. Whether the *P. siculus* populations in Turkey same or different from European populations? If the answer is different, subspecies or species' positions are discussed. However, if it is same, “how and where the species goes into Turkey” are discussed. To learn the answers of the questions, a phylogenetic study should be performed for the *P. siculus* in Turkish population.

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